

Tyler Roberts

Senior Software Engineer - Android/Java/Kotlin

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SUMMARY

Senior Android Engineer with 10+ years building consumer and business mobile products across **SaaS, marketplace, on-demand service, and data-driven platform** environments, specializing in **Android SDK, Java 8–11, Kotlin 1.4–1.9**, and scalable mobile architecture that stays reliable under real production traffic. Strong history of modernizing legacy Java codebases into Kotlin-first apps, improving crash stability, release quality, and delivery speed through **MVVM**, modularization, offline caching, API contract hardening, and CI/CD automation. Known for solving difficult UI, performance, and device-fragmentation issues while shipping features that improved retention, store ratings, and operational supportability.

SKILLS

- **Mobile Development** — **Android SDK, Java 8–11, Kotlin 1.4–1.9**, XML layouts, Jetpack libraries, **AndroidX, Material Design**, multi-module Android architecture, custom views, deep linking, background processing, push notifications, accessibility, app performance tuning
- **Architecture & Patterns** — **MVVM**, MVP, clean architecture, repository pattern, use-case driven design, dependency injection, modularization, state management, navigation architecture, lifecycle-aware components, configuration change handling, offline-first design
- **Jetpack & Core Libraries** — **ViewModel**, LiveData, **Room**, Navigation Component, WorkManager, DataStore, Paging, Retrofit, OkHttp, Gson, Coroutines, Flow, Dagger 2, Hilt, Firebase SDK, Google Maps SDK
- **Backend & Integration** — REST APIs, JSON parsing, WebSocket integration, authentication flows, token refresh, secure session handling, API error modeling, file upload/download, analytics instrumentation, payment and location service integration
- **Data & Storage** — SQLite, **Room**, SharedPreferences, DataStore, caching strategies, sync queues, pagination, local persistence, conflict resolution, relational data modeling, API-to-local schema mapping
- **Testing & Quality** — JUnit, Espresso, Mockito, unit testing, UI testing, integration testing, regression validation, crash analysis, performance profiling, memory leak detection, Play Console monitoring, release checklist automation
- **Cloud, Delivery & Collaboration** — Firebase Crashlytics, Firebase Analytics, Git, GitHub, Bitbucket, Jenkins, GitHub Actions, Gradle, CI/CD pipelines, Play Store release management, Agile delivery, code review, technical documentation, mentoring

EXPERIENCE

Senior Software Engineer

Web Venturez LLC

February 2020 - October 2025

- Led the design and delivery of large Android applications using **Kotlin 1.6–1.9** and **Java 11**, creating scalable mobile flows for booking, account management, notifications, and transaction-heavy customer journeys.
- Migrated legacy feature modules from Java-dominant implementations to a **Kotlin-first** codebase, reducing null-safety defects and improving maintainability by introducing stricter domain models and cleaner UI state handling.
- Implemented **MVVM** with repository and use-case layers, which separated presentation from data access and made new features easier to test, review, and ship across multiple parallel product tracks.
- Resolved a high-impact crash cluster caused by lifecycle timing issues in fragment recreation by refactoring observer registration, tightening state restoration, and standardizing safe navigation patterns across modules.
- Improved cold-start and screen transition performance by optimizing image loading, lazy initialization, RecyclerView diffing, and network request timing, which produced visibly faster launches and smoother navigation on mid-range devices.
- Built resilient offline and retry behavior for unstable mobile networks using local caching, queued sync operations, and idempotent request handling so users could continue key actions even when connectivity degraded.
- Integrated **Room, Retrofit, OkHttp, Coroutines**, and **Flow** into a cleaner data pipeline that simplified pagination, local persistence, and error propagation between repository, domain, and UI layers.

- Added structured crash monitoring with **Firebase Crashlytics** and release-level analytics instrumentation, which helped the team identify regression patterns earlier and cut recurring production incidents by more than **35%**.
- Delivered secure authentication improvements by hardening token refresh behavior, session expiration handling, and edge-case API validation, reducing failed login loops and improving reliability during backend timeouts.
- Supported a gradual UI modernization from older XML-heavy screens into more reusable Android component patterns, enabling faster rollout of new features without destabilizing mature revenue-generating workflows.
- Fixed a difficult memory leak issue tied to adapters and long-lived context references by auditing object lifecycles, removing improper listeners, and validating results with profiler-based regression checks.
- Mentored Android engineers on code review standards, modular design, and production debugging practices, helping the team raise release confidence while shortening turnaround time for hotfixes and feature delivery by **30%**.

Full Stack Engineer

Webiz Digital

July 2017 - January 2020

- Built Android application features in **Java 8** and later **Kotlin 1.4–1.6**, focusing on customer-facing workflows such as onboarding, search, messaging, profile management, and API-driven content delivery.
- Contributed to the migration of older Android support-library code toward **AndroidX** and more maintainable Jetpack patterns, reducing compatibility friction and simplifying long-term library maintenance.
- Developed reusable networking and parsing layers around **Retrofit** and **OkHttp**, improving consistency in API consumption and reducing duplicate request-handling logic across multiple screens and business features.
- Solved a recurring issue where slow API responses caused duplicate submissions by introducing client-side request guards, better loading-state control, and stronger backend response reconciliation logic.
- Enhanced local persistence with **SQLite** and later **Room**, giving the app better caching behavior, faster reloads, and more stable user experiences when reopening content-heavy screens after interruptions.
- Implemented push notification flows, deep links, and background update handling so that marketing campaigns, transactional alerts, and user-return journeys behaved consistently across supported Android devices.
- Reduced crash rates tied to malformed server payloads by adding safer response mapping, defensive null handling, and fallback UI behavior, which improved app stability during partial backend outages.
- Worked closely with design and product teams to refine XML layout behavior for different screen sizes, improving responsiveness and usability across phones with varied resolutions and hardware capabilities.
- Improved Gradle build reliability and release packaging by cleaning dependency conflicts, standardizing build variants, and tightening signing and environment configuration used in staging and production pipelines.
- Introduced unit and UI testing coverage for high-risk flows using **JUnit**, **Espresso**, and **Mockito**, creating more confidence around refactors and helping catch regressions before Play Store submissions.
- Partnered with backend engineers to document API contracts, validate mobile edge cases, and roll out new capabilities in smaller increments, which reduced integration delays and improved sprint predictability by **25%**.

Software Developer

Centric Tech Inc.

October 2013 - June 2017

- Developed early Android business application modules using **Java 7–8**, XML layouts, and traditional Android component patterns, delivering account, reporting, and workflow screens for internal and client-facing use cases.
- Built CRUD-heavy mobile screens connected to REST-style backend services, gaining strong experience in request handling, local validation, list rendering, and practical error messaging for operational users.
- Maintained and enhanced existing Java-based Android code by refactoring oversized activities into cleaner helper and manager classes, which improved readability without forcing risky full rewrites too early.
- Resolved a frustrating issue where screen rotation caused duplicated fragments and broken form state by restructuring activity lifecycle handling and persisting essential UI data more carefully.

- Improved app responsiveness on lower-spec devices by reducing unnecessary view inflation, optimizing adapter updates, and trimming synchronous work that previously blocked the main thread during navigation.
- Implemented safer login and session persistence behavior, helping the app recover better from expired credentials, interrupted requests, and inconsistent backend responses in real production environments.
- Added more reliable file download and upload handling for document-based workflows, including progress feedback and clearer failure messages that reduced repeated support tickets from business users.
- Contributed to early release discipline by organizing build configurations, supporting QA validation, and documenting reproducible bug steps so fixes could be tested consistently before deployment.
- Worked through device-fragmentation issues involving OS differences and manufacturer-specific behavior, adjusting runtime checks and fallback logic to keep core features stable on a wider Android device range.
- Collaborated with senior engineers to prepare the codebase for later modernization by improving package structure, isolating shared utilities, and reducing technical debt in heavily reused application areas.

EDUCATION

Bachelor's degree in Computer Science

Florida State University

2009 – 2013

- Completed a comprehensive CS program emphasizing software engineering fundamentals, data structures, and practical team delivery through multiple hands-on projects and iterative releases.